

Advanced Power BI Practical

Dataset: Use the Sales, Products, Customers, Employees, Region and Date tables provided earlier.

Level: Advanced

Total Marks: 100

Objective

Build an interactive **Sales Performance Dashboard** in Power BI using:

- Star-schema data modeling
 - Dedicated Date table
 - Advanced DAX
 - Time-intelligence
 - Month-over-Month (MoM) analysis
 - Year-to-Date (YTD)
 - Previous Year comparison
 - KPI indicators
 - Dynamic filters and slicers
 - Drill-down analysis
 - Conditional formatting
-

Part A — Advanced Data Model — 15 Marks

1. Create a proper Date table

Create a dedicated Date table covering the complete sales period.

Include:

- Date
- Year
- Month Number
- Month Name
- Quarter
- Year-Month
- Month-Year Sort Column

Example:

`DateTable =`

```
ADDCOLUMNS (
    CALENDAR (
        MIN(Sales[Date]),
        MAX(Sales[Date])
    ),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Year-Month", FORMAT([Date], "YYYY-MM")
)
```

Sort **Month** by **Month Number**.

Mark the table as a **Date Table**.

2. Create relationships

Create the appropriate relationships between:

- Date → Sales
- Products → Sales
- Customers → Sales
- Employees → Sales
- Region → Sales

Use **single-direction filtering** wherever appropriate.

Part B — Basic Measures — 15 Marks

Create these measures:

1. Total Sales

```
Total Sales =
SUM(Sales[SalesAmount])
```

2. Total Cost

```
Total Cost =
SUM(Sales[CostAmount])
```

3. Total Profit

```
Total Profit =  
[Total Sales] - [Total Cost]
```

4. Total Quantity

```
Total Quantity =  
SUM(Sales[Quantity])
```

5. Total Orders

```
Total Orders =  
DISTINCTCOUNT(Sales[SaleID])
```

6. Average Order Value

```
Average Order Value =  
DIVIDE(  
    [Total Sales],  
    [Total Orders]  
)
```

7. Profit Margin %

```
Profit Margin % =  
DIVIDE(  
    [Total Profit],  
    [Total Sales],  
    0  
)
```

Part C — Time Intelligence & MoM Analysis — 25 Marks

This is the **main advanced** section.

1. Previous Month Sales

Create a measure to calculate sales for the previous month.

```
Previous Month Sales =  
CALCULATE(  
    [Total Sales],  
    DATEADD(  
        'DateTable'[Date],  
        -1,  
        MONTH  
    )  
)
```

)

2. Month-over-Month Change

Calculate the difference between current-month and previous-month sales.

```
MoM Change =  
[Total Sales] - [Previous Month Sales]
```

3. MoM Growth %

```
MoM Growth % =  
DIVIDE(  
    [MoM Change],  
    [Previous Month Sales],  
    0  
)
```

4. Previous Year Sales

```
Previous Year Sales =  
CALCULATE(  
    [Total Sales],  
    SAMEPERIODLASTYEAR(  
        'DateTable'[Date]  
    )  
)
```

5. YoY Growth %

```
YoY Growth % =  
DIVIDE(  
    [Total Sales] - [Previous Year Sales],  
    [Previous Year Sales],  
    0  
)
```

6. YTD Sales

```
YTD Sales =  
TOTALYTD(  
    [Total Sales],  
    'DateTable'[Date]  
)
```

7. YTD Profit

```
YTD Profit =  
TOTALYTD(  
    [Total Profit],  
    'DateTable'[Date]  
)
```

8. YTD Previous Year Sales

```
YTD Previous Year Sales =  
CALCULATE(  
    [YTD Sales],  
    SAMEPERIODLASTYEAR(  
        'DateTable'[Date]  
    )  
)
```

Part D — Advanced KPI Measures — 15 Marks

Create KPI measures for:

1. Sales Target

Create a target measure.

For example:

```
Sales Target =  
[Previous Month Sales] * 1.10
```

This represents a target of **10% higher than the previous month**.

2. Target Achievement %

```
Target Achievement % =  
DIVIDE(  
    [Total Sales],  
    [Sales Target],
```

```
) 0
```

3. Sales Status

Create a measure that displays:

- "Above Target" if achievement is $\geq 100\%$
- "Below Target" otherwise.

```
Sales Status =  
IF(  
    [Target Achievement %] >= 1,  
    "Above Target",  
    "Below Target"  
)
```

4. MoM Status

Create a measure that displays:

- "Growth" when MoM Growth is positive
 - "Decline" when MoM Growth is negative
 - "No Change" when it is zero.
-

Part E — Dashboard Visualizations — 15 Marks

Create a professional dashboard containing the following.

1. KPI Cards

Create cards for:

- Total Sales
- Total Profit
- Profit Margin %
- Total Orders
- MoM Growth %
- YoY Growth %

2. Sales Trend

Create a **Line Chart**:

X-axis:

DateTable[Year-Month]

Y-axis:

- Total Sales
- Previous Month Sales

Show the current and previous period for comparison.

3. MoM Analysis

Create a **Column Chart** showing:

- Year-Month
- MoM Change

This should clearly show months with sales growth and decline.

4. Sales vs Target

Create a **Line and Column Chart** showing:

- Total Sales
- Sales Target

This should allow management to compare actual performance against target.

5. Sales by Region

Create a bar chart showing:

Region → Total Sales

6. Profit by Product

Create a bar chart showing:

Product → Total Profit

Sort from highest to lowest profit.

Part F — Slicers & Filters — 10 Marks

Add the following slicers:

Slicer 1

Year

Slicer 2

Month

Slicer 3

Region

Slicer 4

Category

Slicer 5

Customer Segment

Slicer 6

Employee

All major visuals and KPI cards should respond to the slicers.

Part G — Conditional Formatting

Apply conditional formatting to the **MoM Growth %**.

Use:

- Positive growth → positive indicator
- Negative growth → negative indicator
- Zero → neutral indicator

Also apply conditional formatting to the **Sales vs Target** visual.

Part H — Advanced Summary Table

Create a **Matrix** with:

Rows:

Year
↓
Quarter
↓
Month

Values:

- Total Sales
- Previous Month Sales
- MoM Change
- MoM Growth %
- Total Profit
- Profit Margin %
- YTD Sales

The matrix should support **drill-down** from Year → Quarter → Month.

Final Dashboard Layout

SALES PERFORMANCE DASHBOARD					
Total Sales Profit Margin Orders MoM % YoY %					
Year	Month	Region	Category	Segment	Employee
SALES TREND & PREVIOUS MONTH Line Chart					
MoM CHANGE Column Chart			SALES VS TARGET Combo Chart		
SALES BY REGION Bar Chart			PROFIT BY PRODUCT Bar Chart		
MONTHLY PERFORMANCE MATRIX					
Year → Quarter → Month					
Sales PM Sales MoM MoM % Profit YTD					